

How the Proof Works

Thomas Birnie

For the global three-dimensional Navier–Stokes regularity problem, the fluid obeys

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \quad \nabla \cdot u = 0, \quad u(\cdot, 0) = u_0.$$

A strain in the flow pulls a vortex tube along its axis. As the vortex stretches, incompressibility forces it to contract across its width. Each material parcel keeps its volume, so lengthening the tube narrows its cross-section and the rotating core inside it. That strain also acts on vorticity $\omega = \nabla \times u$. Taking the curl of the equation gives

$$\partial_t \omega + (u \cdot \nabla)\omega = (\omega \cdot \nabla)u + \nu \Delta \omega.$$

The left side follows the moving fluid. When vorticity and strain align, the stretching term is positive: $|\omega|$ rises while the core narrows. Since ω is one spatial derivative of u , the velocity changes more sharply across that core. The gradients rise.

That narrowing brings the nonlinear and viscous parts of the equation into direct competition. Transport can keep pulling the profile into a steeper shape; viscosity spreads sharp differences through $\nu \Delta u$. Scaling shows what happens to this balance when the whole profile is compressed. For $\lambda > 1$, set

$$u_\lambda(x, t) = \lambda u(\lambda x, \lambda^2 t), \quad p_\lambda(x, t) = \lambda^2 p(\lambda x, \lambda^2 t).$$

Space has narrowed by $1/\lambda$, while the corresponding time interval has narrowed by $1/\lambda^2$. Every term in the equation acquires a factor λ^3 :

$$\begin{aligned} \partial_t u_\lambda &= \lambda^3 (\partial_t u)(\lambda x, \lambda^2 t), \\ (u_\lambda \cdot \nabla)u_\lambda &= \lambda^3 ((u \cdot \nabla)u)(\lambda x, \lambda^2 t), \\ \nabla p_\lambda &= \lambda^3 (\nabla p)(\lambda x, \lambda^2 t), \\ \nu \Delta u_\lambda &= \lambda^3 (\nu \Delta u)(\lambda x, \lambda^2 t). \end{aligned}$$

Compression does not give either nonlinear transport or viscosity an extra power of scale. It reproduces their balance inside a narrower region and a shorter interval.

The Sobolev scale records that concentration by

$$\|u_\lambda(\cdot, t)\|_{\dot{H}^s} = \lambda^{s-\frac{1}{2}} \|u(\cdot, \lambda^2 t)\|_{\dot{H}^s}.$$

Below $s = \frac{1}{2}$, the norm falls as the profile concentrates. Above $s = \frac{1}{2}$, it rises. At $s = \frac{1}{2}$, it is unchanged. This is the critical height: narrowing gains or loses no power of scale there. Kinetic energy sits below it. A smaller region can carry steeper gradients while its L^2 cost remains finite, so finite energy alone does not close the concentration route.

The time scaling makes the possible cascade just as sharp. If the k -th stage has spatial width λ^{-k} , its natural duration is λ^{-2k} . Those durations can accumulate:

$$\sum_{k=0}^{\infty} \lambda^{-2k} = \frac{1}{1 - \lambda^{-2}} < \infty.$$

An infinite sequence of narrower, faster stages could then fit below one finite time. This is the Zeno cascade the proof has to rule out.

A finite maximal time can persist only if a continuation norm fails; for the smooth history used here, bounded H^3 control of u would already continue the solution. If $\partial_t^{j+1}u$ stayed integrable up to that time in a fixed Sobolev norm, temporal integration would give $\partial_t^j u$ a finite endpoint. Read backward, each failing response forces the next one out of that integrable control.

In order for a singularity to form, velocity has to blow up.

In order for that to happen, its acceleration would also have to blow up, which means its jerk has to blow up, and so on, and so on:

$$u, \quad \partial_t u, \quad \partial_t^2 u, \quad \partial_t^3 u, \quad \dots$$

Following that sequence upward, one would expect less spatial regularity, not more. A response changing faster in time should look rougher in space. The equation reveals the reverse relation.

The first response is

$$\partial_t u = \nu \Delta u - (u \cdot \nabla)u - \nabla p.$$

The next two are

$$\partial_t^2 u = \nu \Delta \partial_t u - (\partial_t u \cdot \nabla)u - (u \cdot \nabla) \partial_t u - \nabla \partial_t p,$$

and

$$\begin{aligned} \partial_t^3 u = & \nu \Delta \partial_t^2 u - (\partial_t^2 u \cdot \nabla)u \\ & - 2(\partial_t u \cdot \nabla) \partial_t u - (u \cdot \nabla) \partial_t^2 u - \nabla \partial_t^2 p. \end{aligned}$$

The product rule has produced one transport term, then two, then three, with the middle term counted twice. At every finite order the complete binomial pattern is

$$\begin{aligned}\partial_t^{r+1}u &= \nu\Delta\partial_t^r u \\ &\quad - \sum_{a=0}^r \binom{r}{a} (\partial_t^a u \cdot \nabla) \partial_t^{r-a} u - \nabla \partial_t^r p.\end{aligned}$$

Only now is there a pattern worth compressing. Write

$$V_r := \partial_t^r u, \quad Q_r := \partial_t^r p.$$

Then

$$V_{r+1} = \nu\Delta V_r - \sum_{a=0}^r \binom{r}{a} (V_a \cdot \nabla) V_{r-a} - \nabla Q_r,$$

while incompressibility fixes the pressure coordinate through

$$Q_r = R_i R_j \sum_{a=0}^r \binom{r}{a} V_{a,i} V_{r-a,j}.$$

Velocity and pressure travel through the ladder together. In each new temporal response, $\nu\Delta V_r$ reaches two spatial derivatives into the preceding response. Transport and pressure carry the coupled terms, but the reversal is already exposed: rising temporal order is tied to rising spatial order.

The critical energy shows exactly where those higher spatial orders sit. Let $\Lambda = (-\Delta)^{1/2}$ and

$$E_s(t) := \frac{1}{2} \|\Lambda^s u(t)\|_2^2, \quad H(t) := E_{1/2}(t).$$

Pairing the equation at height s gives

$$E'_s = \mathcal{P}_s - 2\nu E_{s+1},$$

where $\mathcal{P}_s$ is the complete nonlinear transfer at that height. At the critical height,

$$H' = \mathcal{P}_{1/2} - 2\nu E_{3/2},$$

$$H'' = \partial_t \mathcal{P}_{1/2} - 2\nu \mathcal{P}_{3/2} + (2\nu)^2 E_{5/2},$$

$$H''' = \partial_t^2 \mathcal{P}_{1/2} - 2\nu \partial_t \mathcal{P}_{3/2} + (2\nu)^2 \mathcal{P}_{5/2} - (2\nu)^3 E_{7/2}.$$

Each step upward in time exposes the next Sobolev height. At order n ,

$$H^{(n)} = (-2\nu)^n E_{n+\frac{1}{2}} + \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{j+\frac{1}{2}}.$$

Here is the reversal. The imagined singular cascade predicts less spatial regularity as its temporal responses accelerate. Viscosity links those responses in the opposite direction: the n -th temporal level exposes the spatial level $n + \frac{1}{2}$. Carry every finite temporal level and every finite interaction term to the terminal time, and every finite Sobolev height is present:

$$\bigcap_{m < \infty} H^m(\mathbb{R}^3) = H^\infty(\mathbb{R}^3) \hookrightarrow C^\infty(\mathbb{R}^3).$$

Velocity, acceleration, jerk, and all later responses have led straight back to smoothness.

The identities reveal the coordinates the proof needs. The finite rectangle theorem supplies their whole-terminal control. Fix any finite temporal depth N and spatial height M . For every $0 \leq n \leq N$, it controls the adjacent pair

$$V_n \in L^2(0, T_*; H^{M+1}), \quad V_{n+1} \in L^2(0, T_*; H^{M-1}),$$

together with its pressure coordinates. Equivalently, the finite rectangle

$$\mathcal{R}_{N,M}(u; T) := \sum_{n=0}^N \left(\|V_n\|_{L^2(0,T;H^{M+1})}^2 + \|V_{n+1}\|_{L^2(0,T;H^{M-1})}^2 \right)$$

satisfies

$$\sup_{0 < T < T_*} \mathcal{R}_{N,M}(u; T) < \infty.$$

The bound may depend on N and M ; it does not depend on the cutoff $T < T_*$. Each rectangle is finite, pressure-complete, and drawn from one fixed-viscosity maximal history.

A larger rectangle retains every coordinate shared with a smaller one. The overlaps agree because they are restrictions of that one history. Their compatible projective family is $\mathcal{I}[u_0]$, and for every finite pair (r, m) it gives

$$u \in H^r(0, T_*; H^m(\mathbb{R}^3)).$$

No infinite-order bound is being demanded. Given a finite coordinate, one chooses a rectangle large enough to contain it. Compatibility puts all such choices into one intersection.

Now take the zeroth temporal row, denoted by $\mathcal{S}_0$. For each finite m ,

$$u \in L^2(0, T_*; H^{m+2}).$$

The pressure-completed interaction and the viscous term then belong to $L^1(0, T_*; H^m)$, so the equation gives

$$u_t \in L^1(0, T_*; H^m), \quad u \in W^{1,1}(0, T_*; H^m) \hookrightarrow C([0, T_*]; H^m).$$

The endpoint values at different heights agree because they are limits of one velocity field. They form one terminal state

$$u_* := u(T_*) \in \bigcap_{m < \infty} H_\sigma^m = H_\sigma^\infty \hookrightarrow C^\infty.$$

At rank r , the pressure formula gives Q_r^* from $V_0^*, \dots, V_r^*$, and the recurrence then gives V_{r+1}^* . Induction produces the full pressure-complete terminal jet. Smooth local theory restarts the equation from u_* ; uniqueness glues that restart onto the original history. A finite maximal time would then have been extended, which is impossible. Hence $T_* = \infty$.

This is not an a priori estimate proof. It is an intersection-to-estimate proof. Viscosity relates the velocity's time derivatives to its spatial Sobolev derivatives. Taken through every finite order, those two directions meet in a common intersection. At that intersection, $H^\infty(\mathbb{R}^3) \hookrightarrow C^\infty(\mathbb{R}^3)$ gives smoothness, and the familiar estimates can be taken from whichever finite Sobolev height they require. The estimates come from the intersection, not the other way around.

For a finite spatial derivative order k and $2 \leq p \leq \infty$, choose

$$m > k + \frac{3}{2} - \frac{3}{p}.$$

That finite coordinate is already present, and Sobolev embedding gives

$$\mathcal{I}[u_0] \longrightarrow C_t H_x^m \longrightarrow L_t^\infty W_x^{k,p}.$$

Choosing $m = 1, 2, 3$ recovers, in succession,

$$\begin{aligned} \mathcal{I}[u_0] &\longrightarrow C_t H_x^1 \longrightarrow L_t^\infty L_x^6, \\ \mathcal{I}[u_0] &\longrightarrow C_t H_x^2 \longrightarrow L_t^\infty L_x^\infty, \\ \mathcal{I}[u_0] &\longrightarrow C_t H_x^3 \longrightarrow L_t^\infty W_x^{1,\infty}. \end{aligned}$$

The proof has not climbed toward one privileged estimate. It has built the intersection from which each finite demand can be read.

In the paper's compact notation, the whole motion is

$$\begin{aligned} u_0 &\longrightarrow \{(V_n, Q_n)\}_{n \geq 0} \longrightarrow \{\mathcal{R}_{N,M}[u_0; T_*]\}_{N,M < \infty} \longrightarrow \mathcal{I}[u_0] \\ &\longrightarrow \mathcal{S}_0 \longrightarrow u_* \in H^\infty \longrightarrow \{(V_n^*, Q_n^*)\}_{n \geq 0} \longrightarrow T_* = \infty. \end{aligned}$$